

Impact of AI on Students

Project Overview

Generative AI tools such as ChatGPT, Gemini, and Copilot have become increasingly integrated into students' academic lives. While these technologies may improve productivity and learning outcomes, concerns have emerged regarding dependency, anxiety, burnout, and long-term knowledge retention.

This project analyzes the impact of AI usage on academic performance and student well-being using a dataset containing 50,000 student records.

Objectives

The main objectives of this analysis are:

- Evaluate whether prompt engineering skills improve academic performance.
 - Analyze the relationship between AI dependency and actual AI usage.
 - Investigate the effects of AI dependency on anxiety and burnout.
 - Explore differences between academic disciplines.
 - Study the potential impact of AI usage on skill retention.
-

Dataset

Dataset: Impact of AI on Students — Academic & Well-being Dataset

Source: Kaggle

Records: 50,000 students

License: CC0 Public Domain

The dataset includes variables related to:

- Academic performance.
- AI usage.
- Traditional study habits.
- Anxiety levels.
- Burnout risk.
- Skill retention.
- Prompt engineering skills.

Tools Used

- Google BigQuery
 - SQL
 - Looker Studio
 - Google Docs
-

Methodology

SQL queries were developed to answer specific research questions.

Several derived variables were created:

- Low, medium, and high AI dependency groups.
- Positive and negative skill retention categories.
- High AI usage versus traditional study preference.

Aggregations such as AVG(), COUNT(), and CASE statements were used to build the final analytical dataset.

Key Findings

1. Traditional Study Remains Dominant

Students spend an average of:

- 8.43 weekly hours using AI.
- 11.21 weekly hours studying traditionally.

Although AI tools are widely adopted, traditional study methods remain the primary learning approach.

2. Prompt Engineering Skills Improve GPA

Students with advanced prompt engineering skills achieved the highest GPA gains.

Skill Level	GPA Increase
Beginner	+0.19
Intermediate	+0.19
Advanced	+0.25

This suggests that effectively interacting with AI tools may positively influence academic performance.

3. AI Dependency Does Not Always Mean Higher AI Usage

One of the most important findings of the project is that perceived AI dependency and actual AI usage are not always aligned.

Some students reporting high levels of dependency still dedicate more hours to traditional studying.

This suggests that psychological dependency and AI usage should be analyzed separately.

4. Major Categories

STEM and Medical students obtained the largest GPA improvements associated with AI usage.

Humanities students showed the smallest gains.

5. Anxiety and Dependency

Students with higher perceived AI dependency tend not to report excessively high levels of exam anxiety.

This may indicate a relationship between AI reliance and academic stress.

6. Skill Retention

Most students maintain positive skill retention regardless of AI usage.

However, heavy AI reliance may be associated with slight decreases in long-term knowledge retention.

Dashboard

An interactive dashboard was developed in Looker Studio to communicate the results through KPIs and visualizations.

The dashboard focuses on:

- GPA improvement.
 - AI dependency.
 - Anxiety levels.
 - Skill retention.
 - Academic performance.
-

Conclusions

Generative AI appears to positively influence academic performance, especially among students with advanced prompt engineering skills.

However, a number of students with greater dependency may experience high levels of anxiety.

The findings also demonstrate that AI dependency does not necessarily imply higher AI usage, highlighting the complexity of students' relationships with generative AI.

Future research should explore the long-term effects of AI usage on learning outcomes and knowledge retention.

Data Source

Nagi Sisiro.

Impact of AI on Students — Academic & Well-being Dataset.

Kaggle.

License: CC0 Public Domain.